

$i_x[8:0]$

$i_y[8:0]$

BOOTH 3-BIT
GROUP
GENERATOR

BOOTH
ENCODER

$o_group_0[2:0]$

$o_group_1[2:0]$

$o_group_2[2:0]$

$o_group_3[2:0]$

$o_group_4[2:0]$

o_sel0

o_sel1

o_sel2

o_sel3

o_sel4

18-BIT BOOTH
PRODUCT GENERATOR

18-BIT BOOTH
PRODUCT GENERATOR

18-BIT BOOTH
PRODUCT GENERATOR

18-BIT BOOTH
PRODUCT GENERATOR

18-BIT BOOTH
PRODUCT GENERATOR

SHIFTER

$o_pp0[17:0]$

$o_pp1[17:0]$

$o_pp2[17:0]$

$o_pp3[17:0]$

$o_pp4[17:0]$

BOOTH SHIFTED PRODUCT

